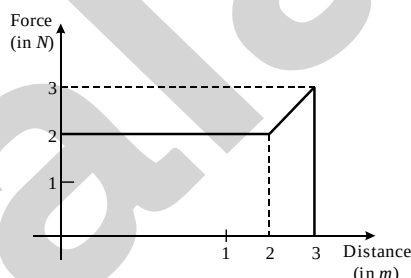


Topic : WEP & CIRCULAR MOTION
Time: 75Min Marking +4 -1
Section - A : MCQs with Single Option Correct

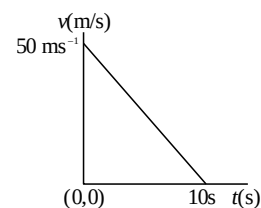
- Two particles of the same mass m are moving in circular orbits because of force, given by $F(r) = \frac{-16}{r} - r^3$. The first particle is at a distance $r = 1$, and the second, at $r = 4$. The best estimate for the ratio of kinetic energies of the first and the second particle is closest to :
 (A) 10^{-1} (B) 6×10^{-2} (C) 6×10^2 (D) 3×10^{-3}
- A body of mass m starts moving from rest along x-axis so that its velocity varies as $v = a\sqrt{s}$ where a is a constant and s is the distance covered by the body. The total work done by all the forces acting on the body in the first t second after the start of the motion is :
 (A) $\frac{1}{8} ma^4 t^2$ (B) $4 ma^4 t^2$ (C) $8 ma^4 t^2$ (D) $\frac{1}{4} ma^4 t^2$
- A small spherical ball is released in a viscous liquid whose density is half that of density of the ball. After attaining terminal speed, ratio of work done by viscous force to work done by gravity on ball during a certain displacement is :
 (A) $\frac{1}{2}$ (B) -2 (C) $-\frac{1}{2}$ (D) 2
- A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3 m is :



- (A) 6.5 J (B) 2.5 J (C) 4 J (D) 5 J
- A ball is thrown upward with an initial velocity V_0 from the surface of the earth. The motion of the ball is affected by a drag force equal to $m\gamma v^2$ (where m is mass of the ball, v is its instantaneous velocity and γ is a constant). Time taken by the ball to rise to its zenith is :
 (A) $\frac{1}{\sqrt{\gamma g}} \sin^{-1} \left(\sqrt{\frac{\gamma}{g}} V_0 \right)$ (B) $\frac{1}{\sqrt{\gamma g}} \tan^{-1} \left(\sqrt{\frac{\gamma}{g}} V_0 \right)$ (C) $\frac{1}{\sqrt{2\gamma g}} \tan^{-1} \left(\sqrt{\frac{2\gamma}{g}} V_0 \right)$ (D) $\frac{1}{\sqrt{\gamma g}} \ln \left(1 + \sqrt{\frac{\gamma}{g}} V_0 \right)$
 - A spring whose unstretched length is l has a force constant k . The spring is cut into two pieces of unstretched lengths l_1 and l_2 where, $l_1 = nl_2$ and n is an integer. The ratio k_1/k_2 of the corresponding force constants, k_1 and k_2 will be :
 (A) $\frac{1}{n^2}$ (B) n^2 (C) $\frac{1}{n}$ (D) n

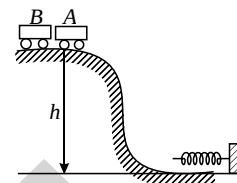
7. Velocity-time graph for a body of mass 10 kg is shown in figure. Work done on the body in first two seconds of the motion is :

(A) -9300 J
(B) 12000 J
(C) -4500 J
(D) -12000 J



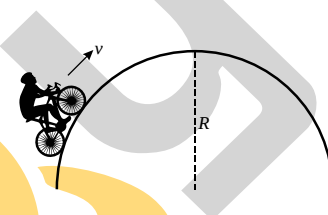
8. Two cars A and B are allowed to slide down a steep slope and are stopped by a large spring. The cars start from rest, it $m_A = 2m_B$ and x_A and x_B are compressions in spring, caused by cars A and B respectively then,

(A) $x_A = \frac{x_B}{2}$
(B) $x_A = x_B$
(C) $x_A = \sqrt{2} x_B$
(D) $x_A = 2x_B$



9. A cyclist is moving on a circular path with a constant speed, the value of normal reaction for the complete path :

(A) Increase
(B) Decrease
(C) Remains constant
(D) First increase then decrease



10. If the road is banked and circular, choose the incorrect option :

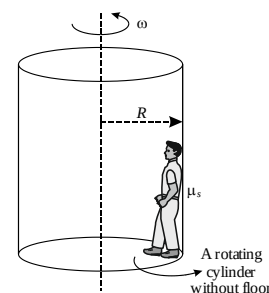
(A) There may be no need of friction for centripetal force
(B) If $N \sin \theta > F_C$, friction must act in the upward direction
(C) If $N \sin \theta < F_C$, friction must act in the downward direction
(D) None of the above

11. Two particles A & B are moving uniformly round concentric circular paths of radii r_A & r_B . If two particles completes one rotation in same time then :

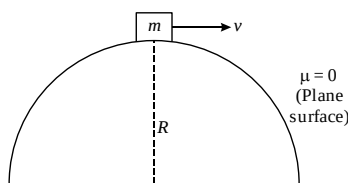
(A) Ratio of their angular velocity is 1
(B) Ratio of their linear velocity is $\frac{r_A}{r_B}$
(C) Ratio of the force required for their motion may not be $\frac{r_A}{r_B}$
(D) All the above

12. A boy of mass m is in touch with the wall of a rotating cylinder. The boy will stay in touch with wall and rotate if :

(A) $\omega \leq \sqrt{\frac{g}{\mu_s R}}$
(B) $\omega \leq \sqrt{\frac{\mu_s R}{g}}$
(C) $\omega \leq \frac{mg}{\mu_s}$
(D) $N \geq \frac{mg}{\mu_s}$



13. A box is placed on a semicircular convex surface. It is given a horizontal speed v . Choose the correct option :



- (A) If $v = \sqrt{Rg}$, the block will leave the surface at the same instant
 (B) If $v > \sqrt{Rg}$, the block will leave the surface after sliding for a very short duration
 (C) If $v < \sqrt{Rg}$, the block will leave the surface after moving some distance along the surface
 (D) None of the above

14. For vertical circular motion as shown in figure, if

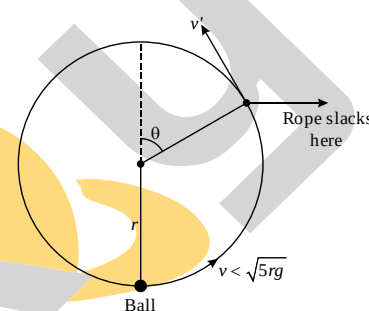
$v_{\text{lowermost point}} < \sqrt{5rg}$, then :

- (A) The rope will slack when $T = mg \cos \theta$

- (B) The rope will slack when $\frac{mv'^2}{r} < mg \cos \theta$

- (C) The rope will slack when $\frac{mv'^2}{r} > mg \cos \theta$

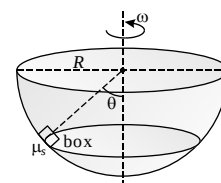
- (D) Rope will leave the circular path at the very next moment when $\frac{mv'^2}{r} = mg \cos \theta$



15. A hemispherical bowl is rotating with an angular speed ω . If there is no need of friction in revolving a box with bowl, which is placed inside the bowl as shown in the figure, then :

- (A) $\cos \theta = \frac{g}{R\omega^2}$ (B) $\sin \theta = \frac{R\omega^2}{g}$

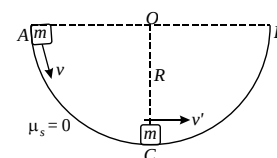
- (C) $\tan \theta = \frac{R}{R\omega^2}$ (D) $\tan \theta = \frac{R\omega}{g}$



16. A box is given a speed v at point A of a hemispherical bowl, then :

- (A) $N_C = 2mg + \frac{mv^2}{R}$ (B) $N_C = 3mg + \frac{mv^2}{R}$

- (C) $N_C = 2mg + \frac{mv'^2}{R}$ (D) $N_C = 3mg + \frac{mv'^2}{R}$



17. A stone of mass M is tied at the end of a string, is moving in a circle of radius R , with a constant angular velocity ω . The total work done on the stone, in any half circle is :

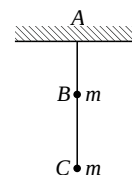
- (A) $\pi MR^2\omega^2$ (B) $2MR^2\omega^2$ (C) $MR^2\omega^2$ (D) 0

18. A body is moving with a constant speed of 5 m/s in a radius of 2 m . The acceleration of the body is :

- (A) 0 (B) 12.5 ms^{-2} (C) 25 m^{-2} (D) None of these

19. A light rod of length l carries two equal masses m , one fixed at the end & other fixed in middle of the rod as shown. The rod can rotate in vertical plane about A. Then horizontal velocity imparted to end C of rod to just deflect the rod upto horizontal position is :

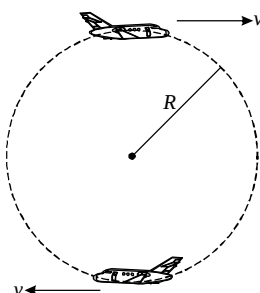
- (A) $\sqrt{\frac{12gl}{5}}$ (B) $\sqrt{3gl}$
 (C) $\sqrt{\frac{16gl}{5}}$ (D) $\sqrt{2gl}$



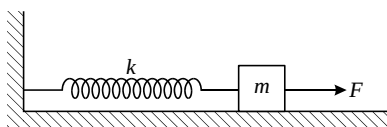
20. A train rounds an unbanked circular road of radius 50m at a speed of 54km/hr . If $g = 10\text{m/sec}^2$, the angle of banking required to prevent wearing out of rails is given by :
 (A) $\theta = \tan^{-1}(0.15)$ (B) $\theta = \tan^{-1}(0.25)$ (C) $\theta = \tan^{-1}(0.35)$ (D) $\theta = \tan^{-1}(0.45)$

Section- B: INTEGER Answer Type Questions

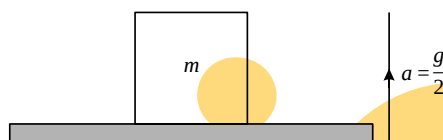
21. A small sphere of mass m is suspended by a light string of length l . It is raised upto the height of suspension with thread fully stretched and released. Then the maximum tension in thread will be nmg . Find the value of 'n':
22. A force acts on a 2 kg object so that its position is given as a function of time as $x = 3t^2 + 5$. What is the work done (joule) by this force in first 5 seconds ?
23. A particle which is experiencing a force, given by $\vec{F} = 3\vec{i} - 12\vec{j}$, undergoes a displacement of $\vec{d} = 4\vec{i}$. If the particle had a kinetic energy of 3J at the beginning of the displacement, what is its kinetic energy (joule) at the end of the displacement ?
24. A body of mass 1 kg falls freely from a height of 100 m on a platform of mass 3 kg which is mounted on a spring having spring constant $k = 1.25 \times 10^6\text{ N/m}$. The body sticks to the platform and the spring's maximum compression is found to be x . Given that $g = 10\text{ ms}^{-2}$, the value of x (cm) will be close to.
25. A uniform cable of mass ' M ' and length ' L ' is placed on a horizontal surface such that its $\left(\frac{1}{n}\right)^{\text{th}}$ part is hanging below the edge of the surface. To lift the hanging part of the cable upto the surface, the work done is $\frac{Mgl}{kn^2}$. Find the value of 'k' :
26. Starting from rest, a particle rotates in a circle of radius $R = \sqrt{2}\text{ m}$ with an angular acceleration $\alpha = \frac{\pi}{4}\text{ rad/sec}^2$. Calculate the magnitude of average velocity of the particle over the time it rotates quarter circle.
27. A jet travelling at a constant speed of $1.20 \times 10^2\text{ m/s}$ executes a vertical loop with a radius of $5.00 \times 10^2\text{ m}$. Find the magnitude of the force (N) of seat on a 70.0 kg pilot at the top of the loop. (Take, $g = 10\text{m/s}^2$) :



28. A block of mass m , lying on a smooth horizontal surface, is attached to a spring (of negligible mass) of spring constant k . The other end of the spring is fixed, as shown in the figure. The block is initially at rest in its equilibrium position. If now the block is pulled with a constant force F , the maximum speed of the block is $\frac{nF}{\sqrt{mk}}$. Find the value of 'n'.



29. A block of mass m is kept on a platform which starts from rest with constant acceleration $g/2$ upward, as shown in figure. Work done by normal reaction on block in time t is $\frac{kmg^2t^2}{8}$. Find the value of k .



30. A ring which can slide along the rod is kept at midpoint of a smooth rod of length L . The rod is rotated with constant angular velocity ω about vertical axis passing through its one end. The ring is released from midpoint. The velocity of the ring, when it just leaves the rod is $\frac{\sqrt{x}}{2}\omega L$. Then x is equal to :

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Section - A : MCQs with Single Option Correct

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|---------|---------|---------|---------|
| 1. (B) | 2. (A) | 3. (C) | 4. (A) |
| 5. (B) | 6. (C) | 7. (C) | 8. (C) |
| 9. (D) | 10. (D) | 11. (D) | 12. (D) |
| 13. (C) | 14. (D) | 15. (A) | 16. (B) |
| 17. (D) | 18. (B) | 19. (A) | 20. (D) |

Section- B: INTEGER Answer Type Questions

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|---------|-----------|------------|---------|
| 21. [3] | 22. [900] | 23. [15] | 24. [4] |
| 25. [2] | 26. [1] | 27. [1316] | 28. [1] |
| 29. [3] | 30. [7] | | |

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